SteadyScreen

A supervised, checkable version of the CDC 30-Second Chair Stand test that runs in a web browser with a laptop camera.

UnivaBio 2026 · Team: Aditya Raut, Neeraj Movva, Sagar Raut, Sathvik Loke · Prototype, not a medical device

The problem

The CDC's 30-Second Chair Stand counts how many times an older adult can stand up from a chair in 30 seconds. It is part of the CDC STEADI falls toolkit. The person running it has to stand next to the participant for safety, time 30 seconds and count every full stand, all at once. Afterwards there is no way to check that count. A camera could do the timing and counting, but a camera counter can be wrong without anyone noticing: a half stand counted as full, a missed stand, or the wrong person tracked.

Camera counting of this test already exists: a University of Alberta iPad app agreed closely with an exercise physiologist (ICC 0.93, PLOS Digital Health, 2026). Its authors note disagreements on borderline stands at the 30-second mark. We are not trying to count better. We built the checking: every stand can be checked against pictures and corrected, and the app withholds the score when unsure.

What SteadyScreen does

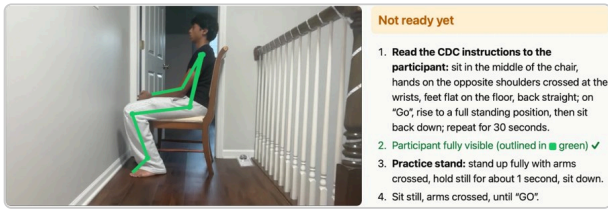
- Guides the setup from the CDC form: a helper beside the participant, a chair with no arm rests, a side-on camera, and the CDC instructions to read out.
- Records one practice stand, which the supervisor confirms from its picture. That posture becomes this person's own standard for a full stand. (The practice stand is our addition, not part of the CDC form.)
- Gives the "Go", times 30 seconds and counts full stands from the camera. The pose model (MediaPipe) runs on the computer; video is not uploaded.
- When it is not sure a rise was a full stand, it marks it "needs your decision" instead of guessing. It applies the CDC rule for a rise that is over halfway up at 30 seconds.
- Keeps temporary still images (needed for a reported camera count) so the supervisor can check each counted stand, confirm the right person was measured, and scan the whole test for a missed stand. Corrections work in both directions and are logged.
- Keeps the supervisor's own count separate. If the two counts disagree, nothing is reported until the supervisor chooses one and gives a reason.
- Shows the CDC comparison table only when every check is complete, and only for ages 60 to 94. Otherwise it lists exactly what is missing.

What we tested

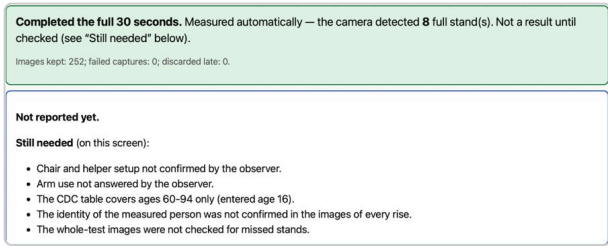
The AI part is MediaPipe Pose Landmarker, a pretrained model that finds body points in each frame; we trained nothing. The counting and checking rules are our own JavaScript, which runs the same way in the browser and in our test suite (342 passing automated tests, plus end-to-end tests with Chrome's simulated camera). We used AI coding assistants while building it, including to review the code many times for ways it could report a wrong count without a warning. Each fix came with a test that failed before it.

On October 5 we ran it in Safari with a laptop camera at home, with a team member as the participant and a helper beside them. On the first try the head and shoulders left the frame when standing. The app was sure of 3 stands and asked us to decide 4 others (3 to 7). Our own count was 8, so the counts disagreed and the app reported no number. After fixing the camera angle, it found 8 full stands on its own, with images for every stand. We kept no separate count of that run, so the 8 is unchecked.

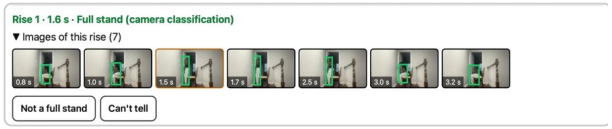
What it is not. SteadyScreen does not diagnose anything or predict falls, and it has not been validated. So far it has only been tried by our team, who are students, not older adults. We have not shown that it is faster or more accurate than a person with a stopwatch. What we built is a way to make a supervised test checkable.



Camera check: the pose model tracks the participant from the side and lists what is still needed before the test can start.



After the test: 8 full stands detected, but "not a result until checked", with the remaining checks listed.



Each stand has its own images; the highlighted one is the top of the stand.

Protocol source: CDC STEADI "Assessment: 30-Second Chair Stand". CDC does not endorse this tool.